

I would really be grateful if you start to build the Weather Clock, that you go to Instructables and say hi.

[Weather Clock Arduino ESP32 : 8 Steps - Instructables](#)

<https://github.com/billbill100/Weather-Clock>

[email billbillgithub@gmail.com](mailto:billbillgithub@gmail.com)

ESP32 Weather Clock User Guide. V1.3 11/07/2026

This project is based on the original open-source code and project details found here.

[ESP32 WiFi Color Display Kit Grande • ThingPulse](#)

Note:- The firmware code used for my version has been heavily modified to make it much simpler to build and use. The original open-source code will work (if you want to try it) but will require editing as all of the user credentials are hard coded.

You will also need to change the TFT pin configuration, either in the code or on the hardware.

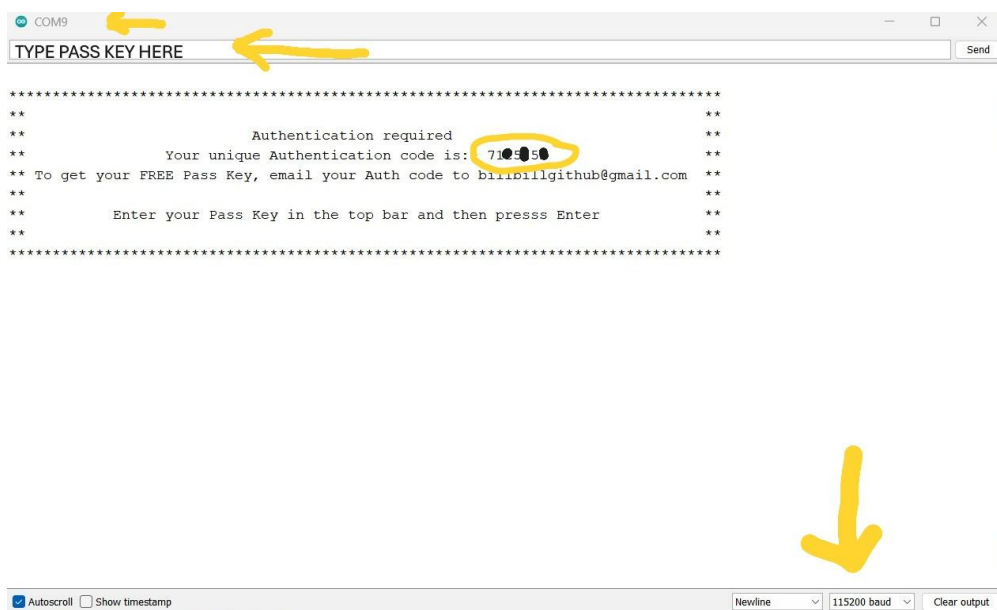
Adding Pass Key

On first use, a one-time Pass Key must be entered. This is issued free of charge and needs entering only once. The reason for the Pass Key is to stop cloning, piracy and third parties selling completed weather clocks using my firmware without permission.

In the 'Arduino IDE Software Load document' you should have already learned how to connect your ESP32 board to the computer, open Serial Monitor and see the ESP32 output to the computer screen.

So again, open Arduino IDE and set it so you can see output from your ESP32 in the Serial Monitor window.

Ensure you drag up the serial Monitor window to make it as large as possible and **press the ESP32 reset button**.



The screen will show your unique Authorisation Code. Email this to billbillgithub@gmail.com your passkey will be sent as soon as possible.

NO NOT progress past this screen without your Pass Key. Failed attempts are recorded, too many will lock you out.

Adding User Details

The user details, as described in the previous document, need to be input to the Weather Clock. Once done, it need never be done again, unless any of the details change, like new WIFI code or moving town.

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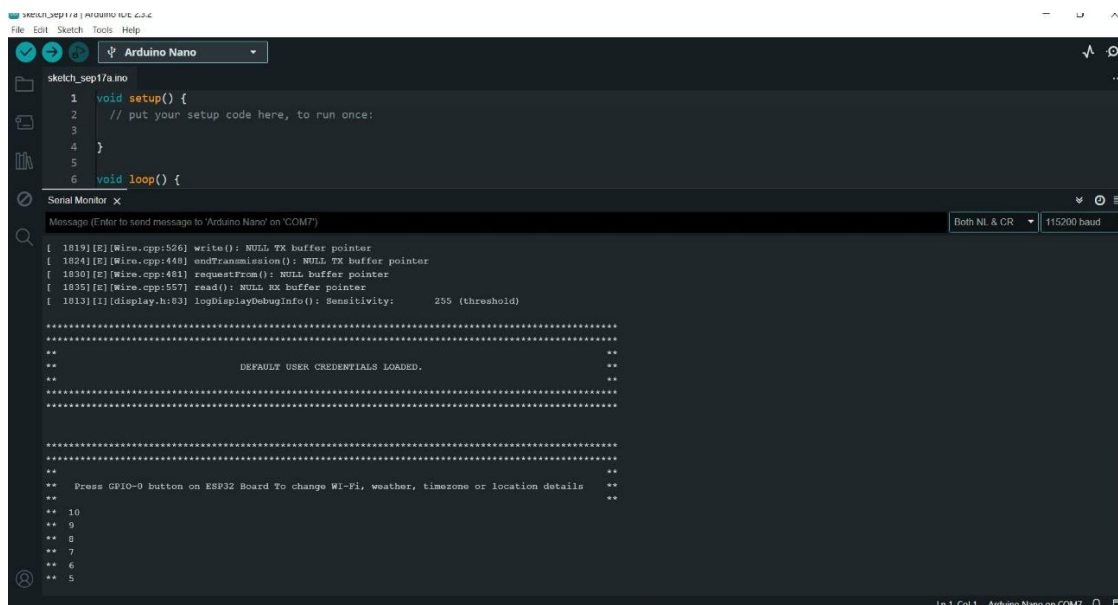
You should see the serial monitor show similar to the below.

The numbers will countdown from 10 to 0.

Within this time, press GPIO 0 button on the ESP32 board or press the Return key on your keyboard.

This will now allow new user data to be input.

Ensure you select 'Both NL & CR' at the end of the black bar on the screen.



```
sketch_sep17a.ino
1 void setup() {
2   // put your setup code here, to run once:
3 }
4
5
6 void loop() {

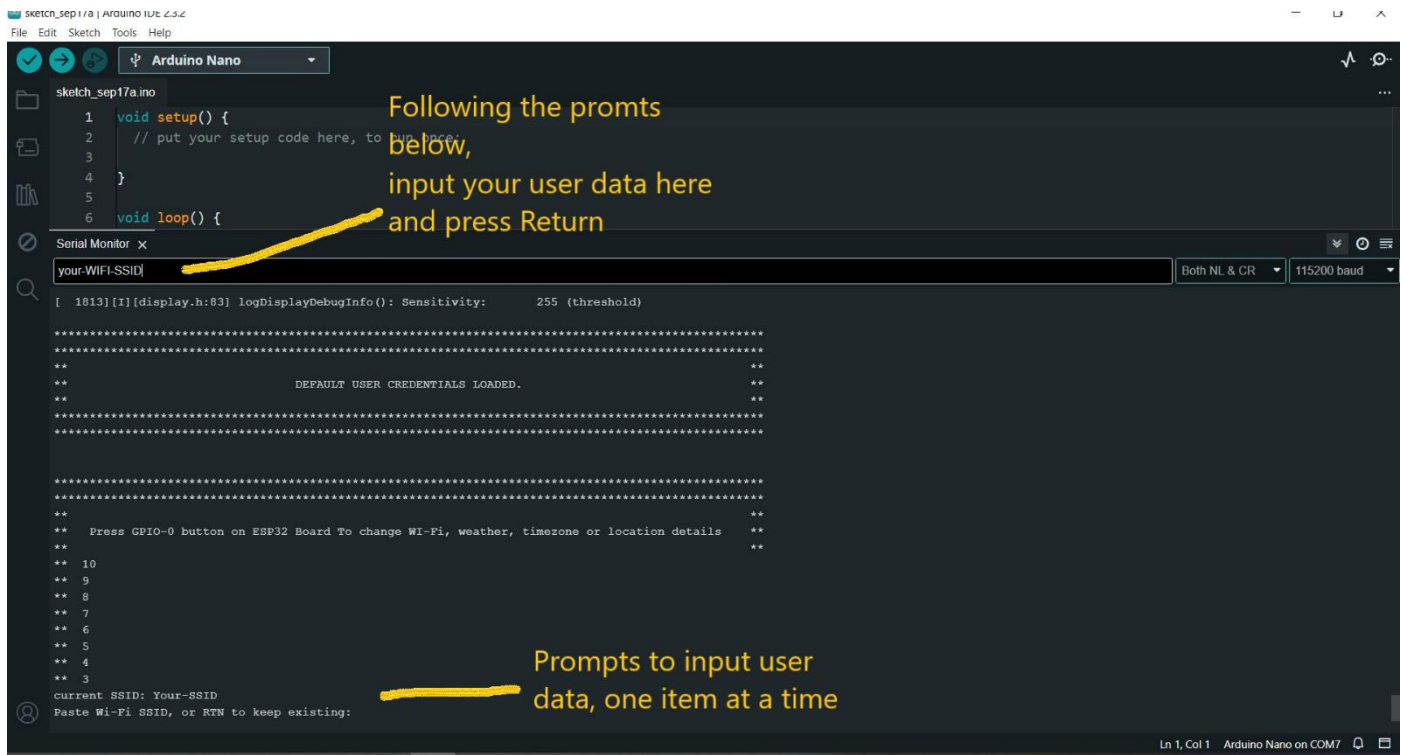
[ 1819][P][Wire.cpp:536] write(): NULL TX buffer pointer
[ 1824][P][Wire.cpp:448] endTransmission(): NULL TX buffer pointer
[ 1830][P][Wire.cpp:481] requestFrom(): NULL buffer pointer
[ 1835][P][Wire.cpp:557] read(): NULL RX buffer pointer
[ 1813][I][display.h:63] logDisplayDebugInfo(): Sensitivity: 255 (threshold)

*****
**                                     **
**          DEFAULT USER CREDENTIALS LOADED.          **
**                                     **
*****

*****
**                                     **
** Press GPIO-0 button on ESP32 Board To change WI-Fi, weather, timezone or location details **
**                                     **
** 10                                     **
** 9                                     **
** 8                                     **
** 7                                     **
** 6                                     **
** 5                                     **

Ln 1, Col 1 Arduino Nano on COM7
```

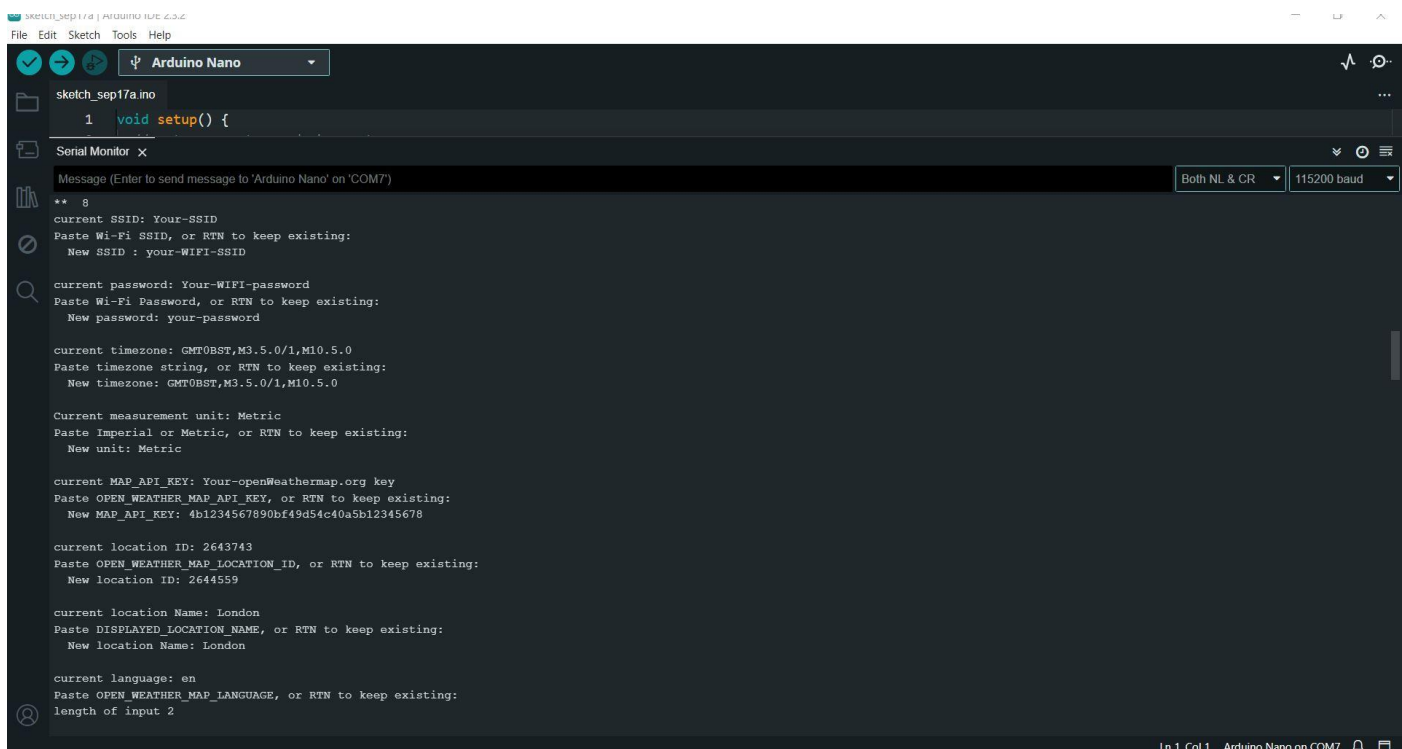
The screen should now look like this



Now copy & paste your WIFI SSID into the black bar and press Return.

The screen will show your newly input data and show the next field to be input. Continue following the prompts and pasting the requested data.

Note: - pressing Return at any prompt without entering data, will cause the current setting to be retained. This is useful, if you only need to change one thing, saving having to copy & paste all data items.



If all of your details are correct, the Weather Station will now show 'Starting WIFI' and then connect, update the Weather to the TFT screen.

Both the TFT screen & Serial Monitor will be helpful in diagnosing issues. For example, if your WIFI details are incorrect, the TFT will stay on the 'Starting WIFI screen & Serial Monitor will keep scrolling, trying to connect.

If the TFT get past the WIFI connection, but the screen shows '?' on the icons, it indicates that the OpenWeather settings are not correct.